Md Shohanoor Rahman

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Research Interests

Trustworthy Machine Learning, Explainable AI (XAI), Generative Adversarial Networks (GANs), Biomedical Image Analysis, and Scalable Software Systems. Seeking an PhD. in CSE to bridge the gap between theoretical deep learning and secure, production-grade diagnostic tools.

Education

- **Rajshahi University of Engineering & Technology (RUET)** Rajshahi, Bangladesh
  Bachelor of Science in Computer Science & Engineering *Jan. 2017 – Aug. 2023*
  - **Academic Standing:** Cumulative GPA: 2.94/4.00 (62.80% Equivalence)
  - **Upper Division Excellence: GPA for Final 60+ Credits (Major Coursework): 3.02/4.00**
  - **Selected Coursework:** Data Structures, Analysis of Algorithms, Operating Systems, Computer Architecture, Distributed Systems, Artificial Intelligence, Database Systems, Digital Signal Processing.

Standardized Test Scores

- **IELTS Academic:** Overall Band 6.5 (L: 7.0, R: 6.5, W: 6.5, S: 6.0) Nov. 2025
- **GRE General Test: Total 312** (Quantitative: 165, Verbal: 147, Analytical Writing: 3.0) July 2025

Experience

- **Brain Station-23 PLC.** Dhaka, Bangladesh
  Software Engineer *March 2023 – Present*

Projects

- **TennantCo. USA:** Tennant Company provides innovative, battery-powered autonomous cleaning solutions integrated into a scalable cloud platform, Optimus, built on Azure. The system enables real-time data collection, aggregation, and insight generation via API-driven architecture, powering portals like Hercules and IRIS.
- **Pocket:** E-Wallet with Admin, Customer, and Merchant Portals using Spring Boot microservices and Angular frontend. Utilized PostgreSQL, Kafka, Redis.
- **AB Direct:** Internet Banking System for AB Bank PLC. with Admin Portal built using Spring Boot microservices, Angular, Kafka, PostgreSQL, Redis.
- **Aerial Satellite Imagery into Map Routes:** Image-to-image translation using Conditional GAN with encoder-decoder and PatchGAN discriminator.
- **Generating Hand-written Images of Digits:** Trained GAN using MNIST dataset to synthesize digit images.
- **Unpaired Image-to-Image Translation:** CycleGAN model to translate horses to zebras and vice versa.
- **Biomedical Image Segmentation:** Implemented U-Net with 23 convolutional layers for segmenting biomedical images.

Research Experience

- **IEEE Publications:** First author of two peer-reviewed papers published in the **2025 28th International Conference on Computer and Information Technology (ICCIT), IEEE**: "Explainable Machine Learning for Chronic Kidney Disease Prediction with SHAP and LIME," DOI: 10.1109/ICCIT68739.2025.11490146; and "Interpretable Multi-Class Obesity Classification using Bayesian Optimization and SHAP Insights," DOI: 10.1109/ICCIT68739.2025.11491728.
- **Ongoing Research — ClinDiffXAI:** Developing *ClinDiffXAI*, a concept-conditioned diffusion framework for controllable and clinically interpretable synthesis of dermoscopic skin-lesion images, with evaluation of concept fidelity and generation quality; manuscript in preparation for *Computer Methods and Programs in Biomedicine* (Elsevier).

- **Undergraduate Thesis: Generative Adversarial Network based Synthetic Data Generation System.** Designed and implemented a CTGAN-based synthetic data generation system , focusing on improving data availability for deep learning applications.

Certifications

- **Deep Learning & GANs (DeepLearning.AI)**
  - Generative Adversarial Networks (GANs) Specialization
  - Build Better Generative Adversarial Networks (GANs)
  - Build Basic Generative Adversarial Networks (GANs)
  - Apply Generative Adversarial Networks (GANs)
  - Neural Networks and Deep Learning
  - Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
  - Convolutional Neural Networks
  - Sequence Models
- **Programming & Data Foundations**
  - Programming for Everybody (Getting Started with Python) – University of Michigan
  - Foundations: Data, Data, Everywhere – Google

Programming Skills

- **Languages:** Python,Java, TypeScript, C/C++, SQL
- **Technologies:** PyTorch, Tensorflow, LaTeX

Management Skills

- JIRA
- Confluence
- Git